

JISHNU JANARDANAN

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EDUCATION

B.S. in Computer Engineering | Illinois Institute of Technology | Chicago, IL

December 2025

Computer Science & Engineering Coursework: Data Structures and Algorithms, Machine Learning, Systems Programming, Application Software Design, Operating Systems, Object-Oriented Programming, Probability and Statistics, Mobile Application Development, Computer Organization and Design, Digital Computers and Computing, Database Organization, Introduction to VLSI Design, Engineering Electronics

CERTIFICATIONS & PUBLICATIONS

IBM Z Xplore – Concepts & Advanced | IBM (Credly Verified) | December 2025

ThinkFast: A Versatile Trivia System for Education, Collaboration, and Communication – Under Review, IEEE EIT 2026 | Paper ID: 121

SKILLS

Programming Languages: Python, Java, JavaScript, Dart, Bash, HTML/CSS, SQL, C, C++, Matlab

Frameworks & Libraries: React, Flutter, Node.js, Spring Boot, OpenMP, GraphQL, JUnit

Databases: PostgreSQL, MySQL, DynamoDB, Redis, Firebase

AI/ML & Performance: Neural Network Design (MLP, CNN), Tensor Optimization (NCHW/NHWC), OpenMP Parallelization, Cache Optimization

Cloud & DevOps: AWS (Lambda, EC2, API Gateway, S3, RDS, CloudFront), Docker, Git, CI/CD (GitHub Actions), MQTT, Linux/Ubuntu

Distributed Systems & System Design: CAP Theorem, Database Sharding & Replication, Caching Strategies, Message Queues (Pub/Sub), Load Balancing, API Gateway patterns, Microservices Architecture, Horizontal Scaling, Fault Tolerance, Rate Limiting

Concepts: REST APIs, TCP/IP, Multi-Threaded Programming, Unit Testing

WORK EXPERIENCE

Backend Developer Intern | FoodCLUB | United Kingdom (Remote)

January 2024 – May 2024

- Maintained 99% uptime for user authentication and data retrieval APIs in a live production environment, as measured by production monitoring dashboards, by developing Node.js backend services on AWS EC2
- Reduced response latency by 30% across core user-facing endpoints by implementing Redis caching to optimize REST API performance in a distributed systems architecture
- Troubleshooted and resolved backend issues through systematic code reviews and log analysis, improving product quality and reducing production incidents
- Standardized request/response formats and error handling for frontend integration by writing technical documentation for all REST API endpoints

TECHNICAL PROJECTS

SneakersForLess | React, AWS Lambda, DynamoDB, Firebase | sneakersforless.org | Production Application

Spring 2026

- Built a sneaker price comparison engine that aggregates and normalizes listings across multiple retailers, surfacing the lowest available price per model and size to end users
- Designed a serverless backend on AWS Lambda, API Gateway, and DynamoDB to handle search queries and persist user watchlists, with Firebase Authentication for OAuth login
- Shipped a React frontend on S3 + CloudFront and automated releases with a GitHub Actions CI/CD pipeline, cutting deployment time by 80%

Old Is Gold – Senior Fitness Platform | React, AWS Lambda, DynamoDB, CloudFront | oldisgold.fit | Production Application

Spring 2026

- Designed personalized workout plans and progress analytics tailored for senior users, accounting for mobility constraints and recovery patterns not handled by mainstream fitness apps
- Built a nutrition tracking module backed by a 50+ item food database with per-serving macro calculations and daily intake summaries
- Implemented the backend as Lambda functions behind API Gateway with DynamoDB persistence, sustaining sub-200ms response times under load testing

IoT Smart Device Hub – Device Management System | Java, Spring Boot, MQTT, REST APIs, JUnit | Illinois Institute of Technology

Spring 2025

- Implemented a Spring Boot backend with pub/sub messaging and the observer pattern, enabling real-time device state synchronization across HTTP and MQTT protocols
- Validated system reliability by building a JUnit test suite with JaCoCo coverage analysis, delivering 94% instruction coverage and 100% test pass rate across 30+ test cases
- Designed 5 RESTful endpoints with CRUD operations and null-guard handling, enabling bulk device control with fan-out commands across plug groups

Neural Network Inference Engine | C++, OpenMP, MNIST | Illinois Institute of Technology

Fall 2024

- Optimized memory access patterns by switching to NCHW/NHWC tensor layouts, improving cache hit rate by 8% and saving 7ms per inference
- Executed MLP and CNN models end-to-end with pre-trained weights, achieving 97.5% classification accuracy on MNIST
- Refactored 300 lines of C++ using OpenMP parallelization, eliminating performance bottlenecks in core computational layers

Think Fast – Embedded Trivia Game | Python, Raspberry Pi, Arduino, UART Serial | Illinois Institute of Technology

Spring 2025

- Integrated Arduino with Raspberry Pi via UART Serial communication for real-time game state management, delivering sub-100ms LED feedback response times
- Resolved all false trigger inputs by implementing software debouncing algorithms, synchronizing audio playback with hardware events

Chicago Transit Authority (CTA) Database System | Python, SQL, MySQL, PostgreSQL | Illinois Institute of Technology

Summer 2024

- Built a relational database with 6-8 entities and referential integrity constraints, ensuring data consistency across transit operations
- Authored complex SQL queries using advanced window functions and OLAP analytics, supporting scalable reporting across diverse business scenarios

INVOLVEMENT

IIT Motorsports (FSAE)

Jan 2024 – May 2024

- Researched quad inverter hardware architecture and authored decision-matrix analysis informing power switch selection for the team's Formula SAE electric race car